

University of Dar es Salaam, College of ICT
IS364: Principles of Systems Security
Project: OSINT Reconnaissance Assessment on UDSM

Assignment Type: Individual Practical Assignment

Submission: PDF Report

Deadline: 16 May 2026

Top 5 will be awarded cash prizes

Objective

The objective of this assignment is to assess students' ability to perform ethical Open-Source Intelligence (OSINT) reconnaissance against a real-world target using publicly available information only. Students are required to conduct passive reconnaissance on the domain: udsu.ac.tz and produce a professional reconnaissance report using a structure similar to industry bug bounty reports.

Important Ethical Scope

This assignment is STRICTLY LIMITED to passive OSINT activities only.

You are allowed to:

- Use search engines
- Use public DNS/WHOIS information
- Use Certificate Transparency logs
- Use Google Dorking
- Use Shodan/Censys and other lookups
- Review publicly accessible metadata
- Investigate exposed documents indexed publicly
- Analyse publicly visible technologies

You are NOT allowed to:

- Perform active scanning (e.g., Nmap)
- Attempt authentication/login
- Use brute force techniques
- Exploit vulnerabilities
- Modify systems/data
- Use Burp Suite against the target
- Conduct phishing/social engineering
- Access restricted/admin panels
- Download sensitive data not intended for public access

Violation of these rules will result in automatic failure of the assignment.

Project Tasks

Perform an OSINT assessment against the target domain and investigate the following areas:

Part A — Domain & Infrastructure Reconnaissance

Identify:

- Domain registration information
- DNS records
- Subdomains
- Public-facing services
- Hosting/cloud providers
- Email infrastructure (MX records)
- SSL/TLS certificates

Part B — Technology Stack Identification

Identify technologies used by the university, such as:

- CMS/frameworks
- Web servers
- Analytics tools
- CDN/WAF technologies
- Email providers
- Learning Management Systems
- Third-party platforms

Part C — Public Document & Metadata Analysis

Use Google Dorking techniques to identify:

- Public PDFs
- DOCX/XLSX/PPT files
- Metadata leakage
- Usernames/emails
- Internal naming conventions
- Software versions
- File paths

You must explain:

- What was discovered
- Why it matters from a security perspective

Part D — Exposure & Risk Analysis

Identify any publicly visible security concerns such as:

- Exposed development systems
- Old portals
- Open directories
- Misconfigured services
- Information disclosure
- Leaked credentials (ONLY through public breach lookup services)
- Publicly exposed sensitive files

Do NOT attempt exploitation.

Part E — Defensive Recommendations

For every finding:

- Explain the risk
- Describe potential attacker abuse
- Provide mitigation recommendations

Required Report Structure

Your report MUST contain the following sections.

1. Title Page

2. Executive Summary

Briefly explain:

- What OSINT was conducted
- Major findings
- Overall security posture observed

3. Scope

Clearly define:

- Target domain
- Allowed activities
- Limitations of the assessment

4. Methodology

Describe:

- Reconnaissance process
- Tools used
- Search techniques
- Validation methods

Include screenshots.

5. Findings

For EACH finding use the following structure:

Finding Title

Severity

Choose:

- Informational
- Low
- Medium
- High
- Critical

Description

Explain the issue clearly.

Evidence

Include:

- Screenshots
- Search results
- URLs
- Metadata
- Tool outputs

Security Impact

Explain how an attacker could abuse the information.

Recommendation

Provide mitigation steps.

6. Overall Risk Assessment

Discuss:

- Common patterns observed
- Information exposure trends
- Overall attack surface

7. Conclusion

Summarise:

- Key lessons learned
- Importance of OSINT in cybersecurity
- Defensive implications

8. References

Include:

- Tools used
- Public sources
- Articles
- Documentation

Use IEEE or APA referencing style.

Submission Requirements

- Submit as PDF
- Professional formatting required
- Include screenshots and evidence
- Use clear section numbering

Important Reminder

This assignment is intended to teach ethical reconnaissance and defensive thinking. The goal is NOT to “hack” the university. The goal is to understand:

- how attackers gather intelligence,
- how organisations unintentionally expose information,
- and how defenders can reduce attack surface exposure.